

Why I chose this particular dataset:

eMBB (enhanced Mobile Broadband):

- mimics consumer internet traffic (video streaming, heavy downloads, web browsing)
- contains large fluctuations in packet sizes, throughput (Bytes), and durations, this makes it intuitive to spot logical relationships

link to my excel workbook

[eMBB_fromOpen.xlsx](#)

resource of original embb

<https://figshare.com/articles/dataset/5G-SliciNdd/24446515>

=COUNTA(A:A) to see how big our sample is, around 6000 rows, very manageable for excel.
52 cols

First thing to be aware of. Check the type of data with TYPE(). I had issues with the way I imported the csv in Excel online and most of it was saved as text. Had to find ways to correct that.

Also formatted it as table for easier calculations.

Used the table filters to search for empty/missing values and do a first inspection of the columns.

Removed protocol=tcp, stos=dTos=0, same as sDsb=dDsb=cs0, scrgap=DstGap=0, state=con, sVId=dVId=NA, predicted = 1:eMBB. Also dur=runtime. Down to columns but still a lot of them

Check text columns (type =2) if they need to trim whitespace (=TRIM()), or to be standardized (with =PROPER() or =UPPER()), but it seems to be a mostly clean dataset.

First column seems to be an arbitrary index.

CLEANING TARGETS: remove any column that seems irrelevant and deal with missing values.

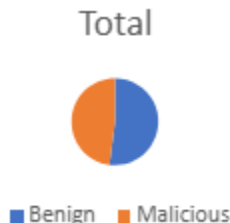
No Duplicates were found, also COUNTBLANK(Table1) =0, so yes, very clean dataset. Or maybe the blanks were filled with zeros, we don't know yet.

Checking for values below zero: =COUNTIF(A5:A5812;"<0") none.

Therefore the challenge of this dataset is not the cleaning part, at least at first glance. It's the 40 or so columns that need to be reduced to a number comfortable for visuals and analysis.

Relatively balanced dataset:

Label	Count of Label
Benign	3023
Malicious	2785
Grand Tot	5808



PART 2

AckDat column dropped since there is no difference in the average between ben-mal.

Label	Average of SynAck	Average of AckDat	Average of TcpRtt	StdDevp of AckDat
Benign	0,016014512	0,019896122	0,035895051	0,006155816
Malicious	0,003487182	0,02037522	0,023830554	0,004934744

Checked zero-variance columns and general exploration with :
st.dev, min, max, average, median, skewness.

This is a large dataset with many columns so for now I will focus on one part of it which is:

Timing

Timing features describe how long a flow lasts and how fast it responds. Attacks often show abnormal timing — very short flows, unusual patterns, or delayed responses.

Dur: total duration of the flow in seconds. Attacks are often abnormally short or abnormally long.

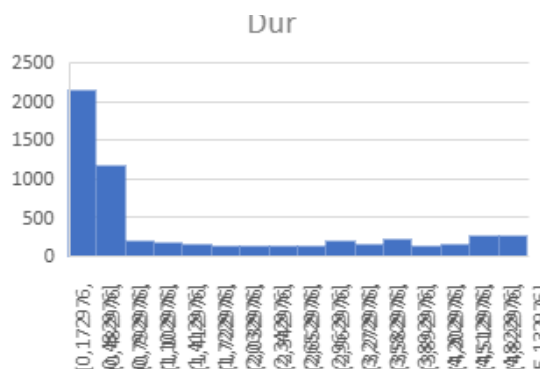
TcpRtt: TCP Round-Trip Time measures connection response speed. Floods and scans respond differently than normal traffic.

SynAck: ratio of SYN to ACK packets, reflects handshake behavior.

Three columns, three angles: how long, how fast, how it greets.

1. DURATION

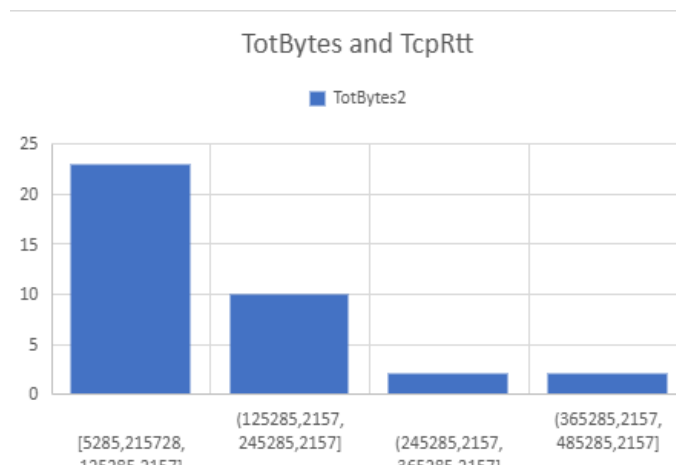
Dur distribution: **right-skewed**. Most flows are very short, with a long tail of longer ones. This means the vast majority of network connections open and close in fractions of a second, while only a tiny fraction last for a long time.



Label	Average of Dur	Max of Dur	Min of Dur	StdDevp of Dur
Benign	2,402246508	4,999976	0,184718	1,670323017
Malicious	0,663681055	4,844445	0,172976	0,819023669

Comparing by label, benign flows show higher variance (std 1.67) than malicious flows (std 0.82), suggesting attack traffic is more uniform in duration. Also malicious flows are 3.6 times shorter on average.

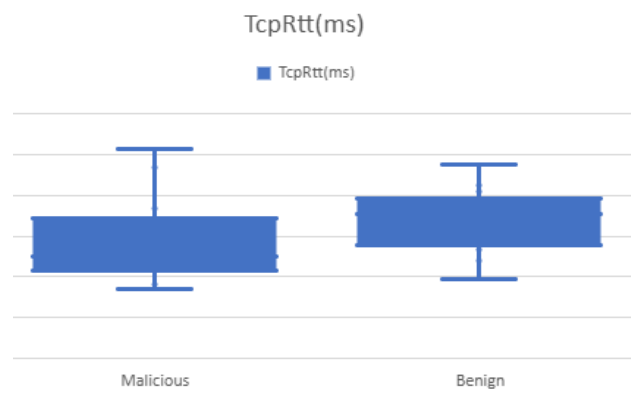
2. RTT



Plotting Total Bytes against TCP Round-Trip Time shows an inverse pattern: higher byte volumes concentrate at low RTT values. This is consistent with normal network behavior : faster connections transfer more data.

Label	Average of TcpRtt(ms)	Max of TcpRtt(ms)	Min of TcpRtt(ms)
Benign	35,89505061	69,788	0
Malicious	23,83055368	52,864	13,926

Malicious tcprtt starts after 13,926 ms stops at 52.864, while it is also lower on average. This narrow, low band separates malicious from benign traffic and makes TcpRtt a useful detection feature.

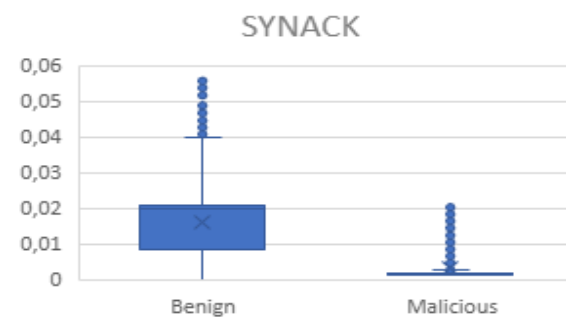


3. SYNACK

Malicious SynAck is **near zero** with almost no variation. Benign is 10 times higher and more spread out. THE box plot shows this visually — malicious is flattened at the bottom.

Label ☐	Average of SynAck	Min of SynAck	Max of SynAck	StdDev of SynAck
Benign	0,02	0,00	0,06	0,01
Malicious	0,00	0,00	0,02	0,00

What this means: SynAck measures the ratio of SYN to ACK packets in a flow; essentially how well the connection completes its handshake. Low value = fewer acknowledgements relative to SYN packets = the receiver isn't replying properly. That's what floods and scans look like, one-sided traffic. Attack flows send requests but receive few acknowledgements, failing to complete a normal handshake. The near-zero SynAck is one of the clearest separators between benign and malicious traffic we have found so far in this dataset.



Thank you for watching!!!